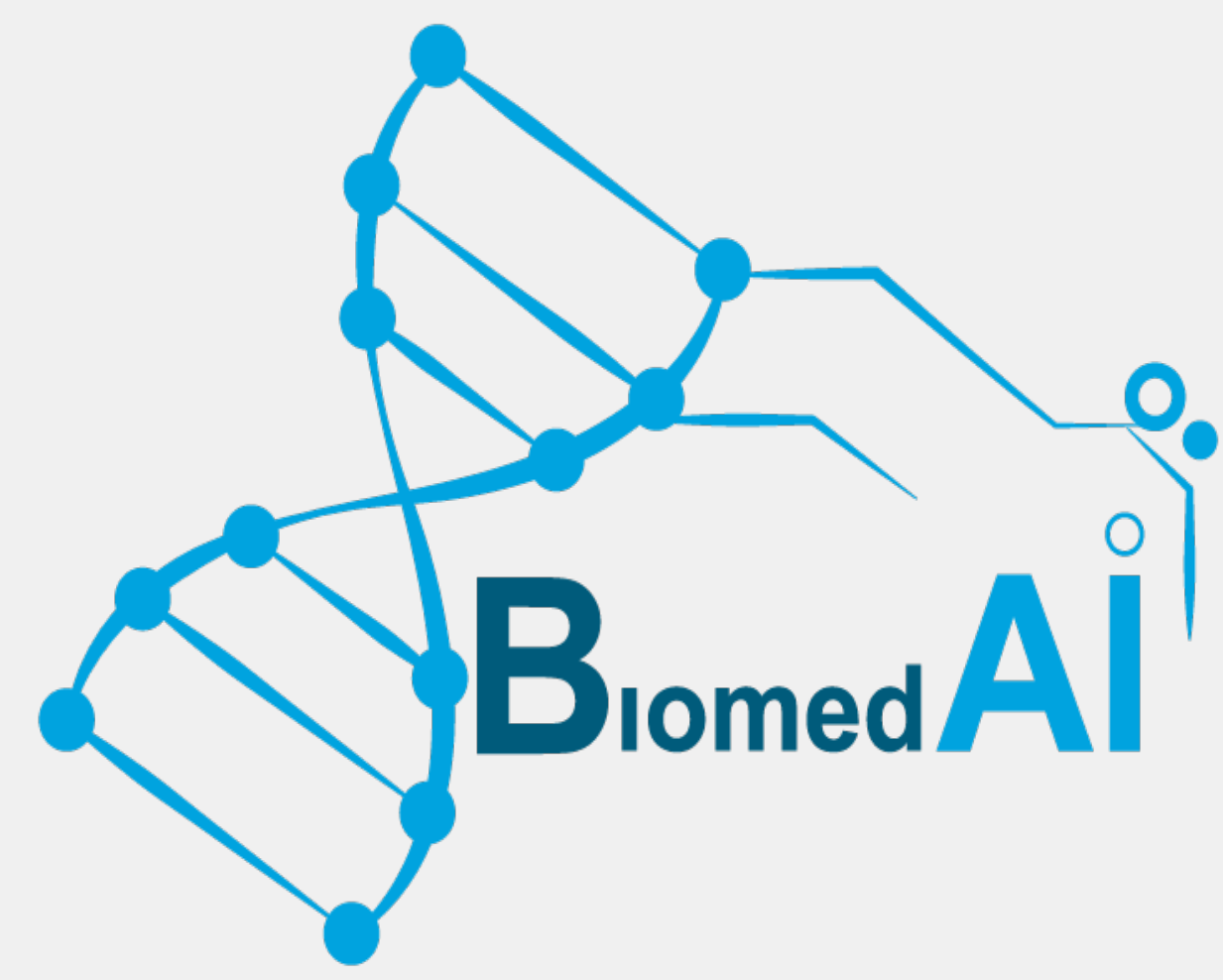




Nonparametric Bayesian **Multi-Facet** Clustering for Longitudinal Data

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Motivation for Multi-facet Clustering

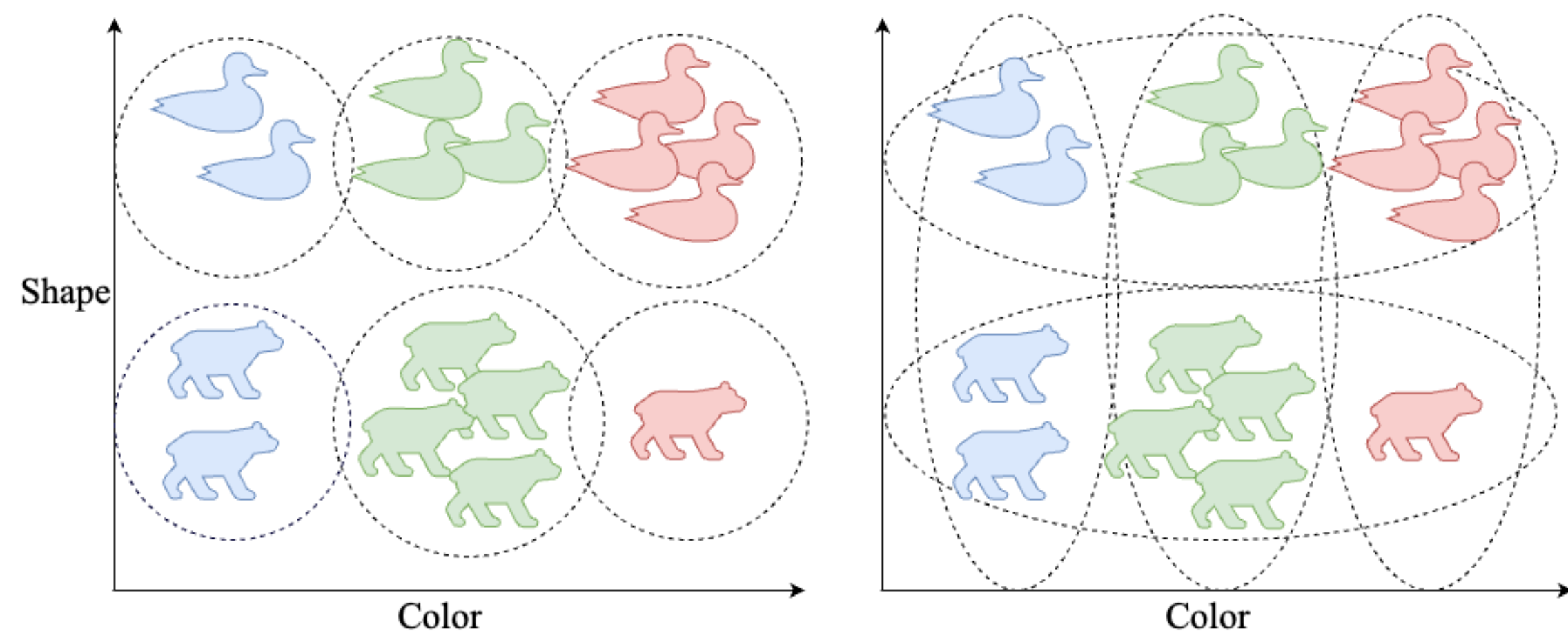


Figure 1. Single-facet versus multi-facet clustering.

- **Single-facet clustering** partitions objects by accumulating contributions from all facets/characteristics, e.g., blue-duck toys, red-bear toys etc. This **lacks interpretability** in providing guidance on which characteristic drives the clustering.
- High-dimensional and complex data structures in many clustering applications may reveal multiple interesting clustering structures, each corresponding to a particular characteristic, or **facet**, e.g., shape or color.
- **Multi-facet clustering** [1] clusters objects based on different characteristics separately but simultaneously, e.g., blue, red and green toys, and duck and bear toys, i.e., each representing different facets.

Multi-facet Mixture Model with Dirichlet Process Priors

The **multi-facet mixture model (MMM)**, is described as

$$\sum_{k_1=1}^{K_1} \cdots \sum_{k_F=1}^{K_F} \pi_{k_1}^{(1)} \cdots \pi_{k_F}^{(F)} p(\mathbf{y} | \boldsymbol{\theta}_{k_1}^{(1)}, \dots, \boldsymbol{\theta}_{k_F}^{(F)})$$

where $f = 1, \dots, F$ are F facets of the mixture component and we assume independence among these facets apriori (plate diagram in Figure 2).

- Specific focus on time-series data, i.e., $\mathbf{y}(\mathbf{t})$
- Facets are defined by the user, e.g., bias, seasonality or trend.
- Each facet has its own mixture model.
- Stick-breaking construction for infinite mixtures (Dirichlet process prior).
- Variational Bayes for scalable inference on large datasets.

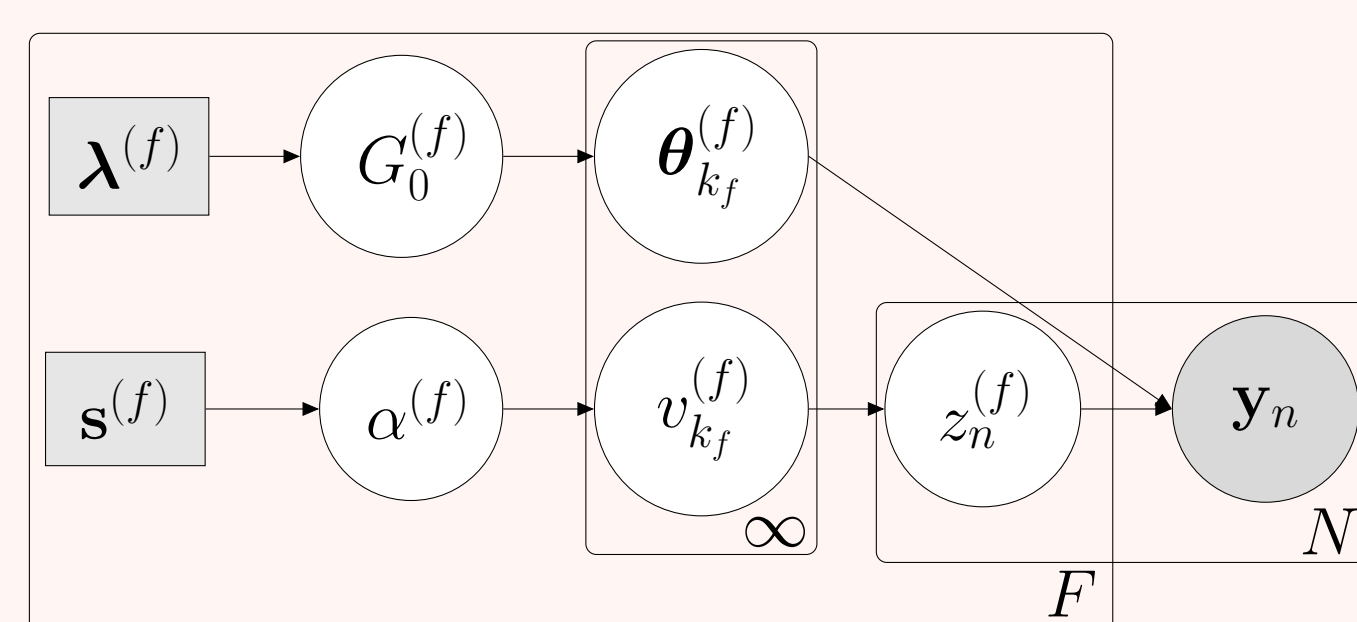


Figure 2. Nonparametric Bayesian multi-facet mixture model.

Two Example Time Series Models

Multi-facet Nonlinear Growth Model

The likelihood of the n -th time series at time $\mathbf{t}_n^{\text{obs}}$ is described as

$$\mathbf{y}_n^{\text{obs}} | z_n^{(a)} = k_a, z_n^{(b)} = k_b, z_n^{(\tau)} = k_\tau \sim \mathcal{N}_{T_n^{\text{obs}}}(\mathbf{a}_{k_a} + \boldsymbol{\beta}_{k_b} \mathcal{B}(\mathbf{t}_n^{\text{obs}}), \boldsymbol{\tau}_{k_\tau} \mathbf{I})$$

where $\mathcal{B}(\mathbf{t}_n^{\text{obs}}) \in \mathbb{R}^{L \times T_n^{\text{obs}}}$ is the basis matrix generated by evaluating spline basis functions at (potentially different) locations $\mathbf{t}_n^{\text{obs}}$.

- Facets: **value** $\mathbf{a}$, **variation** $\boldsymbol{\beta} = (\beta_1, \dots, \beta_L)$ and **volatility** $\boldsymbol{\tau}$.

Multi-facet Vector Autoregressive Model

The likelihood of the n -th time series (of potentially different lengths) at time t is:

$$\mathbf{y}_{nt} | \mathbf{y}_{n(t-1)}, z_n^{(a)} = k_a, z_n^{(b)} = k_b, z_n^{(\tau)} = k_\tau \sim \mathcal{N}_D(\mathbf{a}_{k_a} + \mathbf{B}_{k_b}(\mathbf{y}_{n(t-1)} - \mathbf{a}_{k_a}), \text{diag}(\boldsymbol{\tau}_{k_\tau}))$$

and we assume the outcome vector at the first time point $\mathbf{y}_{n0} \sim \mathcal{N}_D(\mathbf{a}_{k_a}, \text{diag}(\boldsymbol{\tau}_{k_\tau}))$.

- Facets: **values** $\mathbf{a} \in \mathbb{R}^D$, **interactions** $\mathbf{B} \in \mathbb{R}^{D \times D}$ and **noises** $\boldsymbol{\tau} \in \mathbb{R}_+^D$ [2].

Real World Applications

ELSA income data [3]: 13,002 individuals, age range 50–90, 9 waves and up to 9 time points for each individual with 86.8% missing data.

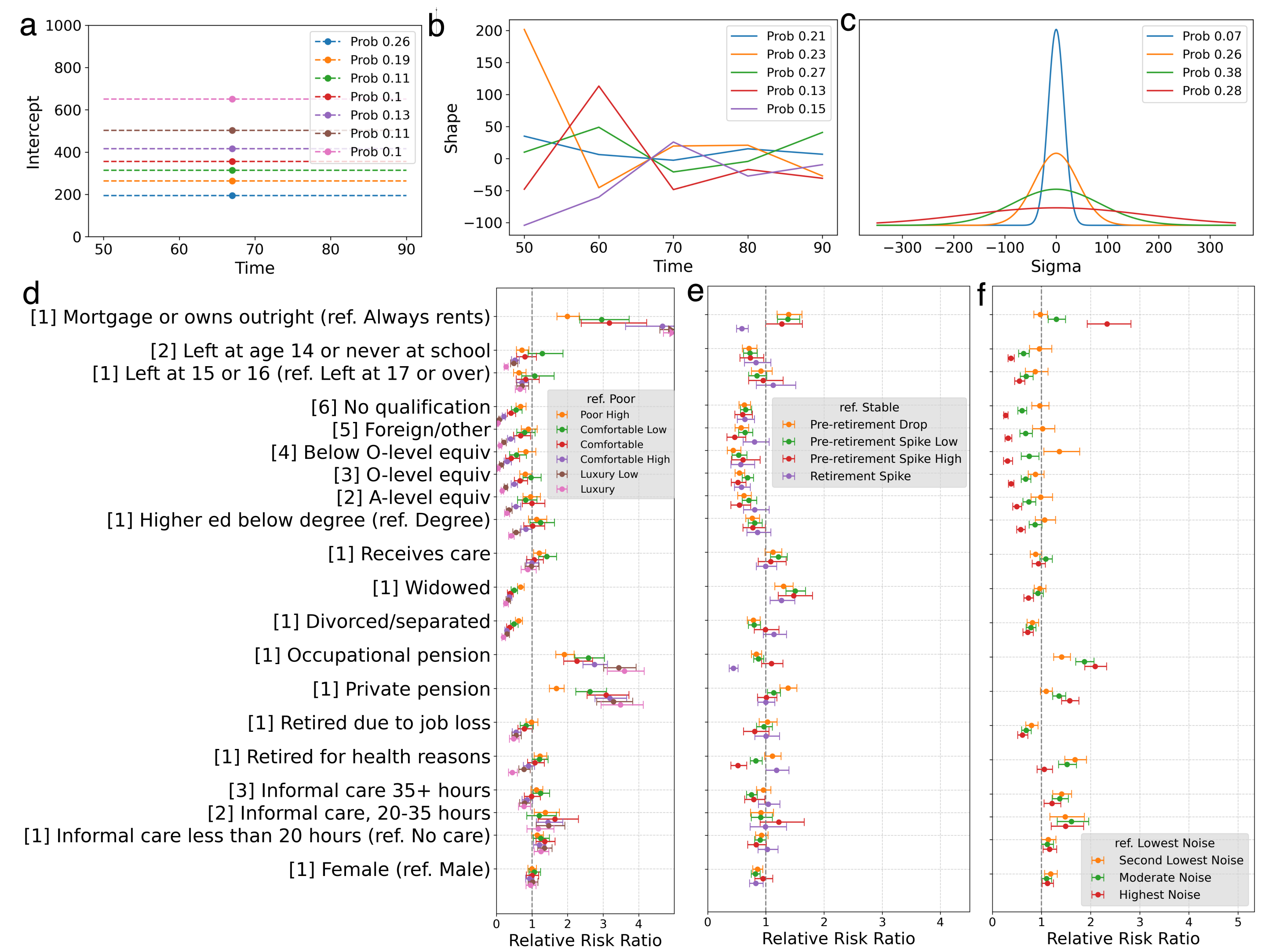


Figure 3. **a)** Seven intercept clusters named “Poor”, “Poor High”, “Comfortable Low”, “Comfortable”, “Comfortable High”, “Luxury Low” and “Luxury” that broadly fall into the categories of “Poor”, “Comfortable” and “Luxury”, at the retirement age of 67. **b)** Five shape clusters, “Stable” income, “Pre-retirement Drop” in income, “Pre-retirement Spike Low” in income, “Pre-retirement Spike High” in income and “Retirement Spike” in income. **c)** Four noise clusters. Relative risk ratios (RRRs) for **d)** intercept **e)** shape and **f)** noise clusters.

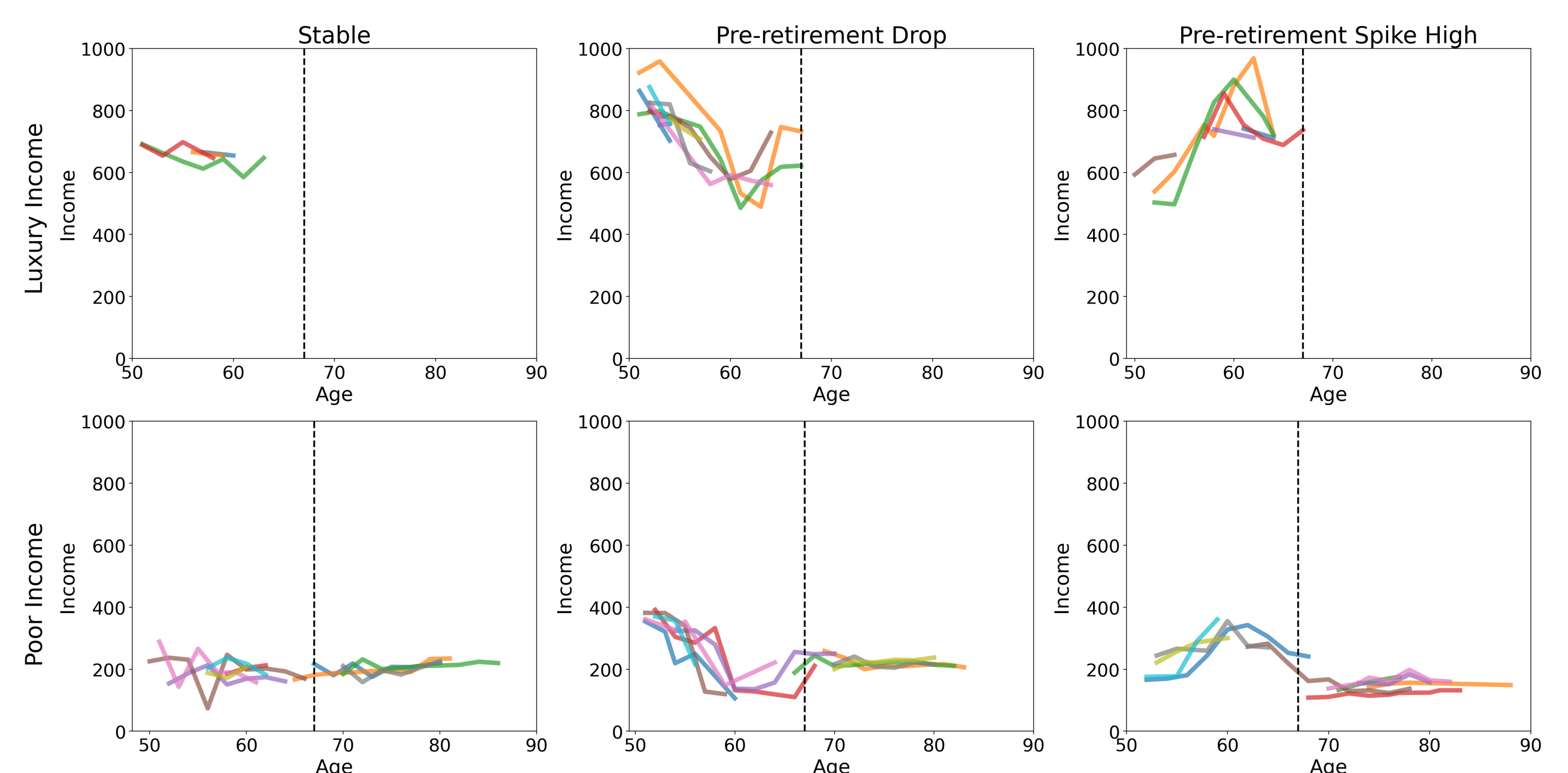


Figure 4. Example income trajectories (with high posterior probabilities) from two facets: income value and income variation. Vertical line shows retirement age.

Discussion and Future Work

Advantages: Disentangling multiple characteristic (**Interpretability**); user-defined facet (**Flexibility**) Variational Inference (**Computational efficiency**); Dirichlet process priors (**automatically learn number of clusters**) and additional **Volatility** facet in income trajectory analysis.

Limitations and Future Directions: Assuming **independence** of facets apriori; **Choice of facets** can vary by specific model designs and interpretation purposes; **Cluster pruning** might require tuning of pruning threshold affecting stability.

References

- [1] Falck, Zhang, Willems, Nicholson, Yau, and Holmes. Multi-Facet Clustering Variational Autoencoders. In *Advances in Neural Information Processing Systems*, volume 34, pages 8676–8690. Curran Associates, Inc., June 2021.
- [2] Kundu and Lukemire. Flexible Bayesian Product Mixture Models for Vector Autoregressions. *J. Mach. Learn. Res.*, 25, 2024.
- [3] Marshall, Nazroo, Tampubolon, and Vanhoutte. Cohort differences in the levels and trajectories of frailty among older people in England. *Journal of Epidemiology & Community Health*, 69(4):316–321, 2015.

